The Polytechnic University
Of The Philippines, Sta. Mesa

OBJECT ORIENTED PROGRAMMING

MIDTERM ASSIGNMENT No. 1:
INTRODUCTION TO JAVA

Submitted by:
Kiel Maverick G. Sumayang
BS Information Technology
2-2N

Date:
February 18, 2026

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Object Oriented Programming

Kiel Maverick G. Samayang

Introduction to Java

1.) What is Java programming language? Discuss its key features and advantages

The Java programming language is a mid-level language that allows the programmer to develop dynamic and complex programs that can easily be shipped through the network without the worry of incompatibility. The Java programming language uses the strength of C++: its capabilities develop and build objects that have their own distinct properties, and the strength of C: its robustness and precision, synthesizing them and adding a list of features: platform independence, security, flexibility, and network compatibility, which overall produces a powerful programming language.

2.) Explain the concept of platform independence in Java. How does Java achieve platform independence?

Platform independence refers to a program having the capability to be implemented on a different operating system or environment without it breaking due to some methods not being made available on another, or simply the architecture is too different from the supported target of the program. Java developed programs are a best example of platform independent programs, as you can ship and implement them on different operating systems or environments — the more common examples are Windows, MacOS, and Linux — without them breaking; this is made possible by the Java Runtime Environment (JRE) which contains the Java virtual machine (JVM). The JVM that is stored inside the JRE allows Java bytecode to be translated into its native counterpart, meaning if you have an environment that primarily runs on C with its specific I/O calls, the bytecode that has the same function as the I/O call is then translated into its equivalent C native code and is then executed.

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5.) Discuss the main components of a Java program and their roles.

In a Java program there are three main components: the Java Development kit (JDK), the Java Runtime Environment (JRE), and the Java Virtual Machine (JVM), these three components work in unison to deliver the full capability and power of Java.

* The Java Virtual Machine (JVM) is a virtual computer hosted inside the Java Runtime Environment (JRE). Its main purpose is to allocate resources for Java to build a specific environment for it to load and execute compiled class files, byte codes. Additionally, it allows the Java program to "adapt" to a specific native environment by allowing Java API calls, so that local java code can be translated by the Java Native Interface (JNI) into its native counterpart.

* The Java Runtime Environment (JRE) is an over encompassing machine that hosts the Java Virtual Machine (JVM). Its overall purpose is to load the class file into objects and push them through the JVM to be executed. JRE and JVM work in tandem when a Java program; they are also responsible for ensuring that a Java program can operate independently from the current platform.

* The Java Development kit (JDK) is the primary environment that allows the programmer to develop a Java program. It provides the programmer the necessary tool to build, test, and run Java programs.

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4.) What are the differences

To discuss the difference between the three key components of a Java program: the JDK, JRE, and JVM, we must first discuss the Java Development Kit (JDK) which contains both the JRE and the JVM. The JDK provides all the necessary tools provided by the JDK is the JRE, which contains the JVM. The Java Runtime Environment (JRE) enables programs to be executed and run it, it loads the necessary class file and passes it through its internal systems such as the JVM. The JVM executes and transforms the bytecode into its corresponding native language which allows the Java program to run on the actual environment.

In summary:

- JDK → An environment for programmers to develop and test Java programs. It contains the JRE which hosts the JVM.

- JRE → One of the primary tool in JDK (in some instance it is separated from the JDK). Loads the class file and executes them, allowing a Java program to run. It contains the JVM.

- JVM → A virtual computer hosted inside the JRE. It allows the standard Java program to run on different types of environment or operating systems by translating them into their corresponding native languages.

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5.) Explain the difference between ~~[illegible]~~ Interpreted and compiled languages. Is Java an interpreted or compiled language.

An interpreted language translates and executes a program line-by-line. This can be helpful as it can catch possible errors in code early. Additionally, it allows programmers to dynamically develop their programs as they can easily retain and change segments of their code while running. On the other hand, a compiled language translates first the entirety of the program before executing it. This helps as it can significantly reduce compile time, enabling programmers to test programs more quickly. However, it's more challenging to debug compiled languages as it doesn't catch recent errors, it instead throws you a collection of errors found after compilation.

The Java programming language is both an interpreted and compiled language. It's a compiled language because all the class files of a program that you want to run is first compiled into bytecodes. It then becomes an interpreted language because it reads the bytecode, or class files line-by-line and processes through the JVM and JIT, allowing the Java program to run.

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